

M12A · MCP showcase

MCP Showcase

Azure + Terraform

What Is MCP?

Model Context Protocol = Toolbox for agents.

Connects Copilot to specialist tools beyond code and files.



AZURE MCP

Query live infrastructure



TERRAFORM MCP

Read state, plan, detect drift

Wire the server. Ask in English. Get specialist output.

Azure MCP — Infrastructure Queries

Connects Copilot to your **Azure subscription**.

Ask questions. Get structured data from Azure Resource Manager APIs.

Example Queries

- "What resource groups do I have?"
- "Show me all storage accounts in West Europe."
- "List my AKS clusters and their node pool configs."

Live queries. Real data. Plain English.

Use for architecture discovery, audits, compliance checks, Terraform planning.

Terraform MCP — State, Plan, Drift

Terraform is DNV's primary IaC tool. Terraform MCP connects Copilot to your infrastructure code.

What Terraform MCP Exposes

- **Read state:** "What resources are deployed in the prod workspace?"
- **Plan changes:** "What would happen if I apply this module?"
- **Detect drift:** "Show me resources that differ from state."
- **Query registry:** "Find Azure verified modules for AKS networking."

The agent now speaks HCL. Your stack's native language.

Demo: Terraform MCP in Action

Prompt: `@terraform what does this module do?`

Copilot reads Terraform module → returns inputs, outputs, provider requirements.

Follow-up: `@terraform show me drift on the prod workspace`

Copilot runs `terraform plan` → surfaces resources that changed outside Terraform.

No CLI. No manual plan. Just English.

Agent suggests: "Run terraform apply to reconcile drift" or "Update state to match live config."

The Combo: Reconcile Azure + Terraform

Azure MCP + Terraform MCP = ask Copilot to reconcile your live state with your code.

AZURE MCP OUTPUT

```
Resource: azurerm_linux_virtual_machine.app
  size: Standard_D4s_v5
  location: westeurope
  tags: { env: "prod", owner: "platform" }
```

TERRAFORM MCP OUTPUT

```
Drift detected:
  ~ size: "Standard_D2s_v5" → "Standard_D4s_v5"
```

```
Copilot suggests:
  • Update Terraform code to match
  • Or revert manual portal change
```

One prompt. Two specialists. Safe remediation.

Next: You Wire the Tools

That's MCP. **Specialist servers. Plain English queries. Infrastructure in your language.**

After Jan Egil's PowerShell MCP demo, you'll return for L4 (Custom Agent Config & MCP Integration):

- Wire Azure MCP, Terraform MCP, PowerShell MCP
- Configure custom instructions
- Turn Copilot into your domain specialist

Then L5: Customize an upstream skill + agent into an ADR Architect for your org.

Come back ready to configure.